

# Spencer Teetaert, PhD Student

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## Education

- 2023 – present     ■ **Ph.D. Student in Aerospace Engineering, University of Toronto** with a focus on state estimation for continuum robots.
- 2018 – 2023     ■ **B.A.Sc. Engineering Science, University of Toronto** in Robotics. GPA: 3.77  
Thesis: *Towards Learning-Based Control for a Tendon-Driven Parallel Continuum Robot*. Available: [https://github.com/spencerteetaert/pcr\\_control/blob/main/thesis/thesis.pdf](https://github.com/spencerteetaert/pcr_control/blob/main/thesis/thesis.pdf)

## Employment History

- 2021 – 2022     ■ **Research Engineer Coop**, Human Machine Interaction Lab, Huawei Canada.  
Developed advanced machine learning computer vision models for eye-gaze and gesture tracking with applications to consumer electronic devices.
- 2019 – 2020     ■ **Industrial Engineering Coop**, Hylife Foods  
Launch an automation overhaul project for a production plant of 2,000+ workers. Built relationships with third party robotics partners and aided in design projects. Lead a cutting blade optimization project.

## TA Positions

- 2024 – present     ■ **ROB498: Robotics Capstone Design**, University of Toronto.
- 2023     ■ **ROB301: Introduction to Robotics**, University of Toronto.

## Research Contributions

### Papers

- 1 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A sliding-window filter for online continuous-time continuum robot state estimation,” 2025, Submitted to Robosoft 2026. arXiv: 2510.26623.
- 2 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A stochastic framework for continuous-time state estimation of continuum robots,” 2025, Submitted to IEEE Transactions on Robotics. arXiv: 2510.01381.
- 3 **S. Teetaert**, W. Zhao, A. Loquercio, *et al.*, “Advancing reproducibility, benchmarks, and education with remote sim2real: Remote simulation to real robot hardware,” *IEEE Robotics & Automation Magazine*, vol. 32, no. 1, pp. 117–123, 2025. DOI: 10.1109/MRA.2025.3527291.
- 4 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “Space-time continuum: Continuous shape and time state estimation for flexible robots,” *ICRA40*, 2024.
- 5 **S. Teetaert**, W. Zhao, N. Xinyuan, *et al.*, “A remote sim2real aerial competition: Fostering reproducibility and solutions’ diversity in robotics challenges,” 2023. arXiv: 2308.16743 [cs.RO].

### Other

- 1 **S. Teetaert**, *Accessible sim2real for benchmarking and hands-on education*, Workshop on Benchmarking, Reproducibility, and Open-Source Code in Controls, IEEE Conference on Decision and Control, 2023.
- 2 Co-organizer, *Safe robot learning competition*, <https://www.dynsyslab.org/iros-2022-safe-robot-learning-competition/>, IROS, 2022.

## Skills

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- Coding     C++, Python, MATLAB,  $\text{\LaTeX}$
- Misc.     3D CAD Design (Autodesk Inventor, Fusion360, SolidWorks)
-  Basic Machining, George Brown College

## Miscellaneous Experience

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### Awards and Achievements

- 2025-26     **Queen Elizabeth II Graduate Scholarship in Science and Technology**
- 2024-25     **Ontario Graduate Scholarship**
- 2022         **NSERC Undergraduate Student Research Award**
- 2018         **Faculty of Applied Science & Engineering Admission Scholarship**
-  **Dr. Dale Iwanoczko Memorial Scholarship**

### Extra-Curriculars and Volunteer

- 2025-present     **University of Toronto Undergraduate Research Students' Association**  
                      • *Graduate Student Mentor*
- 2020-2023       **University of Toronto Robotics Association**  
                      • *President*  
                      • *Project Manager, Autonomous Rover Team*  
                      • *Computer Vision Designer, Autonomous Rover Team*
- 2019-2023       **University of Toronto Engineering Orientation**  
                      • *Subcommittee Chair, Photography*  
                      • *Subcommittee Chair, Retreat*  
                      • *Group Leader*